Revision-Note Review Worksheet for Expert Rater

This booklet contains one student revision note to review. It is a take-home revision note prepared by a teacher for a JSS1 Mathematics class on Three dimensional figures: properties of cubes, cuboids, pyramids and cones. Judge how well it covers the intended objectives and curriculum, follows the teacher's lesson plan, stays on topic and is conceptually correct, and is clear and useful for the students.

Lesson and revision-note context

Subject	Mathematics
Grade	JSS1
Term	Third Term
Week	4
Topic	Three Dimensional Shapes
Subtopic	Three dimensional figures: properties of cubes, cuboids, pyramids and cones
Theme	—
Curriculum objectives	• Identify the properties of cube and cuboids • Identify the properties of pyramids and cones
Previous knowledge	• Students have been taught the differences between cube and cuboid.
Lesson sequence	• Identify the properties of cube and cuboids — the teacher writes the properties of cubes and cuboids on the board; the student take notes in their notebooks • Identify the properties of pyramids and cones — the teacher draws the pyramids and the cone on the board and writes their properties; the student take them down, as in Step II
Instructional materials	• Shapes of cubes, cuboid, Pyramid, cone, cylinder and spheres.
Evaluation	• How many vertices, has a cuboid? • How many edges has a cube? • Draw a cube and cuboid then label. • Draw an hexagonal base pyramid and list the properties. • How many vertices has a tetrahedron.

Lesson objectives

1.	Identify the properties of cube and cuboids
2.	Identify the properties of pyramids and cones

Read the student revision note on the following pages, then complete the rating sheet at the end of this booklet.

STUDENT REVISION NOTE

Topic: Three dimensional figures: properties of cubes, cuboids, pyramids and cones

Class: JSS1 **Subject:** Mathematics **Week:** 4

Section 1: Cubes and Cuboids: Faces, Edges and Vertices

Solid shapes that are not flat are called three-dimensional (3D) shapes because they have length, width and height. Two 3D shapes you meet everywhere are the cube and the cuboid.

Think of a dice, a Rubik's cube or a box of salt — these are all 3D shapes. To describe any 3D shape, you use three special parts:

A face is a flat surface of the shape. When you feel the top of a box, you are touching one of its faces.
An edge is a straight line where two faces meet. Run your finger along the corner of a book and you are tracing an edge.
A vertex (plural vertices) is a corner, where three or more edges meet. The sharp points of a box are its vertices.

Now let us look closely at a cube. A cube is a special 3D shape where every face is a square and every edge is the same length. Because all its faces are equal squares, a cube looks exactly the same no matter which face you look at.

A cube has these properties:

6 faces — all of them squares and all of them the same size.
12 edges — all the same length.
8 vertices — every corner where three edges meet.

A cuboid, sometimes called a rectangular prism or a rectangular box, is like a stretched cube. Its faces are rectangles (though some may be squares), and opposite faces are equal to each other. A shoe box or a book is a cuboid.

A cuboid has these properties:

6 faces — they are rectangles; the two opposite faces are equal to each other.
12 edges — but not all the same length; edges come in sets of four equal ones.
8 vertices.

Notice the important difference: both a cube and a cuboid have 6 faces, 12 edges and 8 vertices. The difference is in their faces. Every face of a cube is a square, while the faces of a cuboid are rectangles (only some of them might be squares).

Worked Example 1: Describing a cube

A student picks up a die. How many faces, edges and vertices does it have, and what shape are its faces?

Solution:
A die is a cube. A cube has 6 faces, 12 edges and 8 vertices. Since every face of a cube is a square, the faces are all squares, and they are all the same size.

Worked Example 2: Cube or cuboid?

Look at these two boxes.

Box A is a salt container with all its sides the same length and square faces.
Box B is a shoe box with rectangular faces, where opposite faces match.

Which one is a cube and which one is a cuboid?

Solution:
Box A has equal edges and square faces, so it is a cube. Box B has rectangular faces where only opposite faces are equal, so it is a cuboid.

Worked Example 3: Counting edges on a cuboid

A cuboid has faces that come in three pairs of equal rectangles. How many edges does it have, and how do the edges group?

Solution:
A cuboid has 12 edges. They are not all equal. There are four edges along the length, four along the width and four along the height. So the 12 edges come in three sets of 4 equal edges.

Practice

1. A Rubik's cube is a cube. How many faces, edges and vertices does it have, and what shape are all its faces?
2. State whether each shape is a cube or a cuboid, and give one reason for your choice.
   (a) A gift box where every face is a rectangle and opposite faces are equal.
   (b) A block where every face is a square and every edge is the same length.
3. A cuboid has 6 faces and 8 vertices. How many edges does it have? Explain whether all these edges are the same length.

Section 2: Pyramids and Cones: Faces, Edges and Vertices

We have looked at cubes and cuboids, where every face is flat and every face is a rectangle or square. Pyramids and cones are different: they rise up to a single point. That point is called the apex (you can also call it the vertex at the top).

<h3>Pyramids</h3>
A pyramid has one flat base, which can be any polygon (a triangle, a square, a pentagon, and so on). Every other face of a pyramid is a triangle, and all those triangles meet together at the apex above the base.

To count the parts of a pyramid, look at the shape of its base. Suppose the base has n sides:

Faces: the base is 1 face, plus one triangular face for each side of the base, so the total is n + 1 faces.
Edges: the base has n edges around its rim, and each of those base corners sends one sloping edge up to the apex, giving another n edges. Total is 2n edges.
Vertices: the base has n corners, plus the apex at the top, giving n + 1 vertices.

For example, a square-based pyramid has a square base, so n = 4. That gives 4 + 1 = 5 faces, 2 × 4 = 8 edges, and 4 + 1 = 5 vertices.

<h3>Cones</h3>
A cone is like a pyramid, but its base is a circle. A cone has:

Faces: one flat circular base and one curved surface that tapers to the apex. That is 2 faces.
Edge: one edge — the circular rim where the base meets the curved surface.
Vertices: just the apex point at the top, so 1 vertex.

Notice the difference: a pyramid's sloping sides are flat triangles, while a cone's sloping surface is curved and smooth, with no corners along the sides.

Example 1 — A triangular-based pyramid
A triangular-based pyramid (also called a tetrahedron) has a triangular base, so n = 3.

Faces: n + 1 = 3 + 1 = 4 faces (the triangular base plus three triangular sloping faces).
Edges: 2n = 2 × 3 = 6 edges.
Vertices: n + 1 = 3 + 1 = 4 vertices.

Example 2 — A cone
A cone has a circular base, so it is not counted with the formula above (a circle has no corners).

Faces: 2 (one circular base and one curved surface).
Edges: 1 (the circular rim around the base).
Vertices: 1 (the apex at the top).

Practice

1. A square-based pyramid has a square base. How many faces, edges, and vertices does it have?
2. A hexagonal-based pyramid has a hexagonal base (a base with 6 sides). Find the number of faces, edges, and vertices of this pyramid.
3. State the number of faces, edges, and vertices of a cone.

Check your work

Attempt every question before you read this page.

Cubes and Cuboids: Faces, Edges and Vertices

1. 6 faces, 12 edges and 8 vertices; all faces are squares.

   A Rubik's cube is a cube, so it has 6 faces, 12 edges and 8 vertices. Because every face of a cube is a square, all 6 faces are squares and they are all the same size.
2. (a) Cuboid — its faces are rectangles with opposite faces equal. (b) Cube — every face is a square and every edge is the same length.

   For (a), the faces are rectangles (not all squares) and only opposite faces match, which is the property of a cuboid. For (b), every face is a square and all edges are equal, which is exactly the property of a cube.
3. 12 edges. They are not all the same length; they come in three sets of 4 equal edges (length, width and height).

   A cuboid, like a cube, has 12 edges. But unlike a cube, the edges are not all equal: there are 4 edges of one length, 4 of another and 4 of a third, matching the three pairs of opposite faces.

Pyramids and Cones: Faces, Edges and Vertices

1. 5 faces, 8 edges, 5 vertices.

   The base is a square, so n = 4. Faces = n + 1 = 4 + 1 = 5. Edges = 2n = 2 × 4 = 8. Vertices = n + 1 = 4 + 1 = 5.
2. 7 faces, 12 edges, 7 vertices.

   The base has n = 6 sides. Faces = n + 1 = 6 + 1 = 7. Edges = 2n = 2 × 6 = 12. Vertices = n + 1 = 6 + 1 = 7.
3. 2 faces, 1 edge, 1 vertex.

   A cone has one flat circular base and one curved surface: that is 2 faces. The circular rim where the base meets the curved surface is its 1 edge. Its only vertex is the apex at the top.

Rating sheet

Mark one score per criterion, then use the notes section for strengths, concerns, and recommendations. The lesson objectives are on the first page for reference.

Revision note ID	cand-7c33da58	**Date**	
Subtopic	Three dimensional figures: properties of cubes, cuboids, pyramids and cones	**Rater**	

> The reference sheet includes the teacher's evaluation questions. A pack that reuses them, reworded or verbatim, may be credited for matching how the lesson was assessed — but reuse on its own does not establish taught-lesson alignment. Rate alignment on the whole of it: the objectives covered, the order taught, what the activities had the students do, the method, and the expected depth.

Criterion (select one score per row)	**1** Poor	**2** Weak	**3** Adequate	**4** Good	**5** Excellent
Objective and curriculum alignment Does the revision note cover the intended lesson objectives and curriculum target?	○	○	○	○	○
Taught-lesson alignment Does it reflect the taught lesson — its sequence, emphasis, activities, and expected depth?	○	○	○	○	○
Scope and conceptual correctness Does it stay within the expected lesson boundary and avoid conceptual/mathematical errors?	○	○	○	○	○
Student-facing clarity and usefulness Can JSS1 learners understand and use it meaningfully as take-home review material?	○	○	○	○	○

Strengths	Concerns	Recommendation

Flag any problems (optional)

Tick all that apply. These are separate from the scores above — leave blank if none apply.

☐	**Misses a required objective** — Skips something the lesson was meant to teach.
☐	**Goes beyond this lesson** — Includes material outside this lesson's scope (e.g. next week's topic).
☐	**Wrong or misleading content** — Contains a mathematical or factual mistake.
☐	**Too hard for JSS1 to follow** — Likely confusing for the students.
☐	**Other:** _____